

Lecture 25 — Simulation and long-run behavior

Wed Oct 21

Last updated: October 8, 2026.

Reading: Simulation of linear dynamics (class notes / Boyd 9.1–9.3 continued). Related objectives: [Lectures](#). [Schedule](#).

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Note: Today we **observe** long-run patterns by simulating. We do **not** yet explain them with eigenvalues — that mystery returns in Week 13.

Learning objectives

By the end of class, students should be able to:

1. Simulate many time steps computationally.
2. Plot state components through time.
3. Recognize stabilization, growth, decay, or oscillation.
4. Conduct a simple what-if simulation.
5. State limitations of a linear dynamical model.

Warm-up

With

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

compute x_1 and x_2 by hand. Guess (no calculator marathon): after many steps, will x_t blow up, shrink to zero, or settle somewhere nonzero?

Fill in

Simulation recipe (any language):

1. Choose A and initial state x_0 .
2. For $t = 0, 1, \dots, T - 1$: set $x_{t+1} = \underline{\hspace{2cm}}$.
3. Store or plot the components of $x_0, x_1, \dots, x_T$ versus time.

Long-run patterns (names only — recognize from plots / tables):

Pattern	What you see
Stabilization	x_t approaches a fixed vector (or settles in magnitude)
Growth	components grow without bound (often roughly geometric)
Decay	components shrink toward $\underline{\hspace{2cm}}$

Pattern	What you see
Oscillation	components bounce / cycle in sign or value

What-if. Change one thing — a parameter inside A , or the initial state x_0 — and re-simulate. Compare the two trajectories.

Limitations of $x_{t+1} = Ax_t$. The model is _____ (no x^2 terms, no products of states); A is assumed constant; time is discrete; there is no noise, capacity limit, or policy change unless you build it into A . Real systems may only match the model for a while.

Worked in class

W1. Many steps on the board / laptop. Let

$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute (or run) x_1 through x_5 . Sketch each component versus t . Does the trajectory look like growth, decay, oscillation, or stabilization?

W2. Recognize the pattern. Without naming any fancy theory, match each matrix to the most plausible long-run story for a generic nonzero x_0 :

(a) $A = \text{diag}(1.2, 0.5)$

(b) $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

(c) $A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$

W3. What-if. Same A as W1, but start from $x_0 = (0, 1)$ instead of $(1, 0)$. Compute x_1 and x_2 . Do the two initial conditions seem headed toward the *same* long-run mix, or different ones?

W4. Limitations. A linear infection model $x_{t+1} = Ax_t$ with fixed A predicts infections forever according to the same rule. Name two real-world reasons this could fail after a few weeks (policy, behavior, medicine, saturation, ...).

Your turn

Y1. Simulation plan. Write 3–4 lines of pseudocode (or Octave/Python if you prefer) that, given A and x_0 , prints $x_0, x_1, \dots, x_{10}$.

Y2. Pattern ID. You simulate $x_{t+1} = Ax_t$ and obtain the first component sequence

$$3, -3, 3, -3, 3, \dots$$

Which pattern name fits best? Give one 2×2 matrix that could produce oscillation like this from $x_0 = (3, 0)$ (many answers OK; one simple choice is enough).

Y3. What-if. Let $A = \text{diag}(0.5, 0.5)$ and $x_0 = (8, 2)$.

(a) Compute x_1 and x_2 by hand.

(b) If instead $x_0 = (80, 20)$, how do x_1, x_2 compare? What limitation / feature of *linearity* does this illustrate?

Y4. Conjecture (no proof needed). For

$$A = \begin{bmatrix} 1.1 & 0 \\ 0 & 0.5 \end{bmatrix},$$

compare long-run behavior starting from $x_0 = (1, 0)$ versus $x_0 = (0, 1)$. Write one sentence you want explained later in the course.

Answers (Your turn)

Y1. Example pseudocode:

```
x ← x0
print x
for t = 1 to 10:
    x ← A x
    print x
```

(Octave: $\mathbf{x} = \mathbf{A}*\mathbf{x}$ in a for loop; Python: $\mathbf{x} = \mathbf{A} @ \mathbf{x}$.)

Y2. **Oscillation.** One simple choice: $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (or $A = -I$, or a 180° rotation), which sends $(3, 0) \mapsto (-3, 0) \mapsto (3, 0) \mapsto \dots$.

Y3. (a) $x_1 = (4, 1)$, $x_2 = (2, 0.5)$. (b) $x_1 = (40, 10)$, $x_2 = (20, 5)$ — exactly $10\times$ the previous trajectory. Linearity scales the whole path with the initial state; the model has no “saturation” that would break that scaling.

Y4. Sample conjecture: from $(1, 0)$ the state grows without bound along the first axis; from $(0, 1)$ it decays to zero — so long-run fate seems to depend on the *direction* of x_0 , not just its size. (Week 13 will explain this with eigenvectors.)